

Pipeline Data Science yang Sudah Diperbaiki

Masalah utama di notebook sebelumnya:

- `X = df.drop('HARGA', axis=1)` masih mengambil kolom teks
- Kolom teks menyebabkan error:

```
could not convert string to float
```

Karena model machine learning hanya bisa membaca angka.

Urutan Yang Benar

1. Import Library

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
```

2. Load Dataset

```
df = pd.read_excel('data.xlsx')
```

3. Cek Data Awal

```
print(df.head())

print(df.info())

print(df.isnull().sum())
```

4. Hapus Kolom Tidak Penting

```
if 'NO' in df.columns:  
    df.drop(columns=['NO'], inplace=True)
```

5. Ambil Hanya Kolom Numerik

INI BAGIAN PALING PENTING.

```
df = df.select_dtypes(include=np.number)
```

Kenapa? Karena dataset rumah biasanya punya kolom teks seperti:

- judul rumah
- lokasi
- deskripsi

Kolom teks menyebabkan error saat scaling atau machine learning.

6. Cek Tipe Data Lagi

```
print(df.dtypes)
```

Pastikan semuanya:

```
int64  
float64
```

7. Handle Missing Value

Karena dataset rumah banyak outlier:

- median lebih aman daripada mean

```
df.fillna(df.median(numeric_only=True), inplace=True)
```

8. Ambil Kolom Numerik

```
num_cols = df.columns
```

9. Visualisasi Outlier

```
plt.figure(figsize=(12,6))

sns.boxplot(data=df[num_cols], orient='h')

plt.title('Boxplot Sebelum Handling Outlier')

plt.show()
```

10. Handling Outlier

Strategi terbaik untuk rumah:

- cap
- bukan remove

Karena rumah mahal masih valid.

```
df_handled = df.copy()

for col in num_cols:

    Q1 = df_handled[col].quantile(0.25)
    Q3 = df_handled[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df_handled[col] = df_handled[col].clip(lower, upper)
```

11. Boxplot Setelah Handling

```
plt.figure(figsize=(12,6))

sns.boxplot(data=df_handled[num_cols], orient='h')

plt.title('Setelah Handling Outlier')

plt.show()
```

12. Heatmap Korelasi

```
plt.figure(figsize=(10,6))

corr = df_handled.corr(numeric_only=True)

sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title('Correlation Heatmap')

plt.show()
```

13. Feature Engineering

```
df_handled['TOTAL_RUANG'] = (
    df_handled['KT'] +
    df_handled['KM']
)
```

14. Log Transform Harga

Sangat disarankan untuk dataset rumah.

```
df_handled['HARGA'] = np.log1p(df_handled['HARGA'])
```

15. Split Data

```
X = df_handled.drop('HARGA', axis=1)

y = df_handled['HARGA']
```

16. Train Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

17. Scaling

```
scaler = MinMaxScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

18. Training Model

```
model = LinearRegression()

model.fit(X_train_scaled, y_train)
```

19. Prediksi

```
y_pred = model.predict(X_test_scaled)
```

20. Evaluasi

```
print('MAE :', mean_absolute_error(y_test, y_pred))  
  
print('R2 Score :', r2_score(y_test, y_pred))
```

Penyebab Error Sebelumnya

Error:

```
could not convert string to float
```

Terjadi karena:

```
X = df.drop('HARGA', axis=1)
```

masih membawa kolom teks.

Solusi utamanya:

```
df = df.select_dtypes(include=np.number)
```

Harus dilakukan SEBELUM:

- scaling
- train_test_split
- machine learning

Saran Model Selanjutnya

Setelah Linear Regression berhasil, coba:

```
from sklearn.ensemble import RandomForestRegressor
```

Karena dataset rumah biasanya lebih cocok dengan Random Forest dibanding Linear Regression.